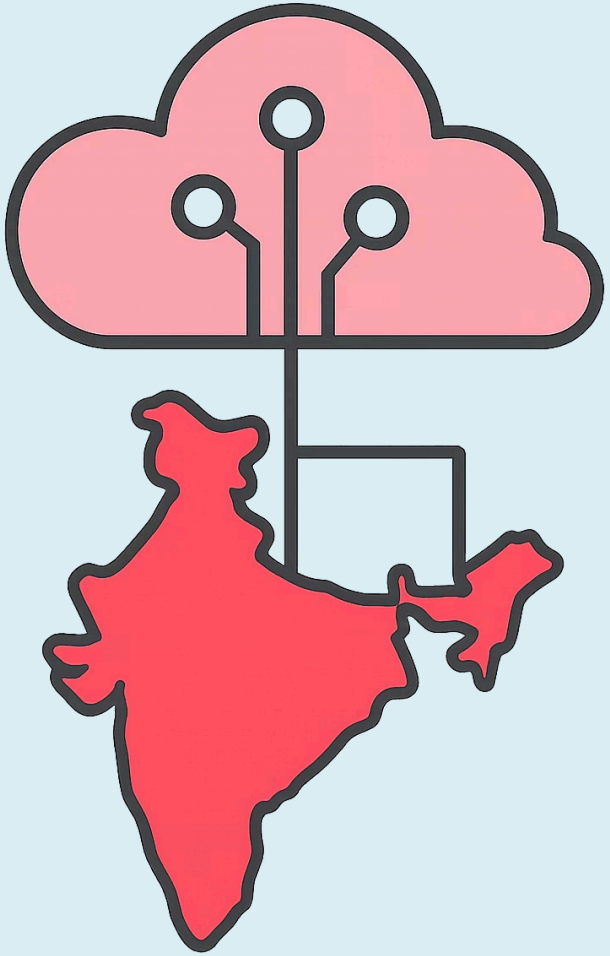

GI CLOUD INITIATIVE

MeghRaj (GI Cloud): A Paradigm Shift in Indian E-Governance

Structural Analysis, Strategic Implementation, and Future Roadmap

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Presented by: Kanishka 002(LE), Lakshmi 004(LE), Lavanya 005(LE), Siddiqua 006(LE), Tanisha 007(LE), Jyoti 057

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HISTORICAL CONTEXT

The Origin of GI Cloud

MeghRaj emerged from a decade-long strategic commitment to national digital infrastructure, evolving from foundational connectivity programs into a unified national cloud platform.

1

2006

National e-Governance Plan (NeGP) establishes core infrastructure: SWAN, SDC, and CSC networks.

2

April 2013

GI Cloud Strategic Direction Paper and Implementation Roadmap formally issued, crystallizing the national cloud vision.

3

February 4, 2014

Former Union Minister **Kapil Sibal** officially launches the National Cloud under the **"MeghRaj"** brand.

4

Administrative Oversight

A dedicated **Task Force** constituted by DeitY to drive strategic direction and implementation.

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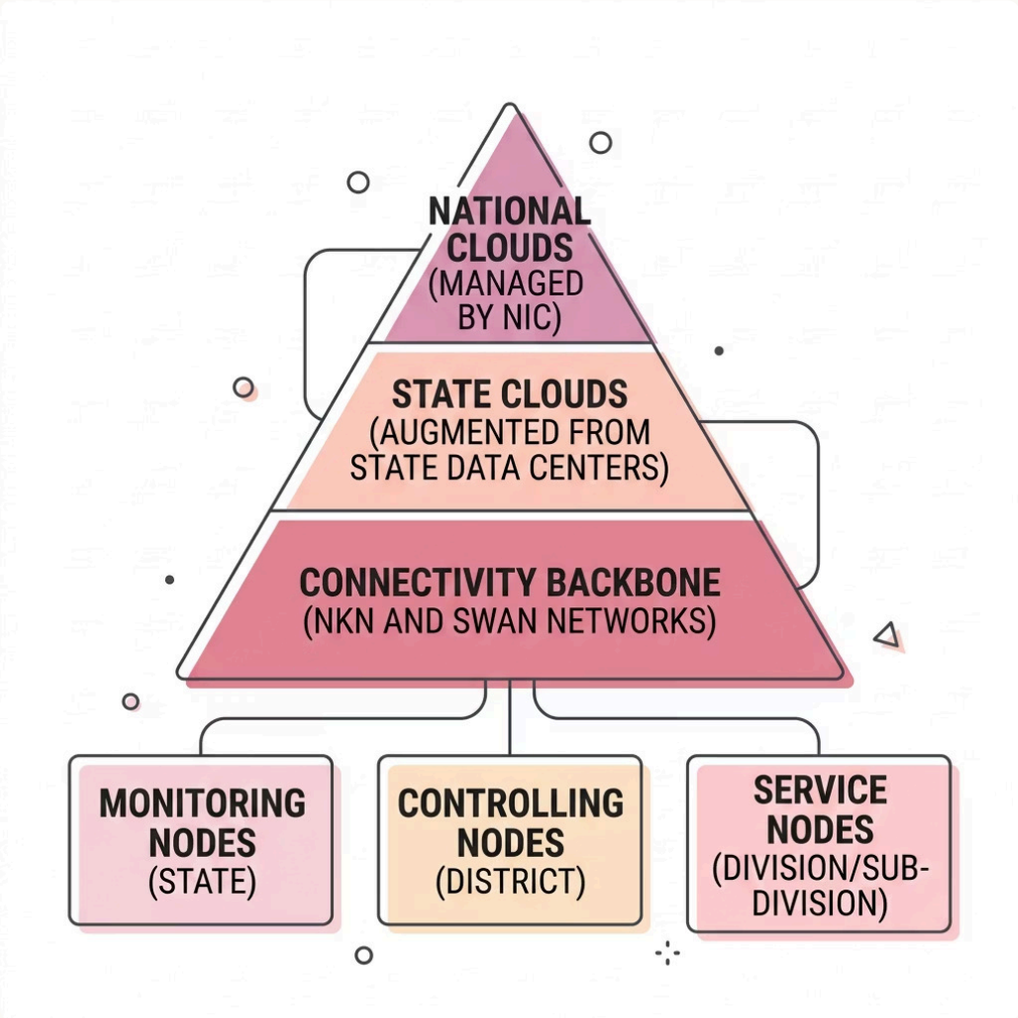
ARCHITECTURE

Structural Mapping & Core Architecture



Three-Tier Node Hierarchy

The architecture mirrors India's administrative structure, ensuring governance at every level:

- **Monitoring Nodes** — State-level oversight and coordination
- **Controlling Nodes** — District-level administration and routing
- **Service Nodes** — Division and sub-division service delivery

Distributed Cloud Environment

Multiple National Clouds (NIC-managed) and State Clouds are integrated via virtualization through the **National Knowledge Network (NKN)** and **State Wide Area Networks (SWAN)**, creating a resilient, federated infrastructure.

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ECOSYSTEM

The GI Cloud Ecosystem — 8 Key Actors

Operational success depends on eight interdependent entities, each fulfilling a distinct governance, technical, or service role.

Cloud Consumer

Government departments, citizens, and industry players utilizing cloud services.

Developer Community

Owners and creators of specialized e-governance applications.

Centre of Excellence

Dedicated bodies for R&D, advisory services, and best practice creation.

Institutional Setup

Administrative departments responsible for monitoring and implementation.

Cloud Provider

Entities maintaining infrastructure and meeting Service Level Agreements (SLAs).

Cloud Auditor

Independent bodies certifying security, privacy, and performance compliance.

Cloud Broker

Managers of service delivery and performance optimization across providers.

Cloud Carrier

Intermediaries providing the transport backbone for cloud data movement.

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USE CASES

Niche Areas & Specialized Implementations

MeghRaj powers domain-specific platforms across critical government sectors, demonstrating the cloud's versatility and reach.

Library Automation

e-Granthalaya 4.0 — A multi-tenant SaaS serving **5,300+ libraries** with UNICODE support for regional scripts.

Railways & Grievances

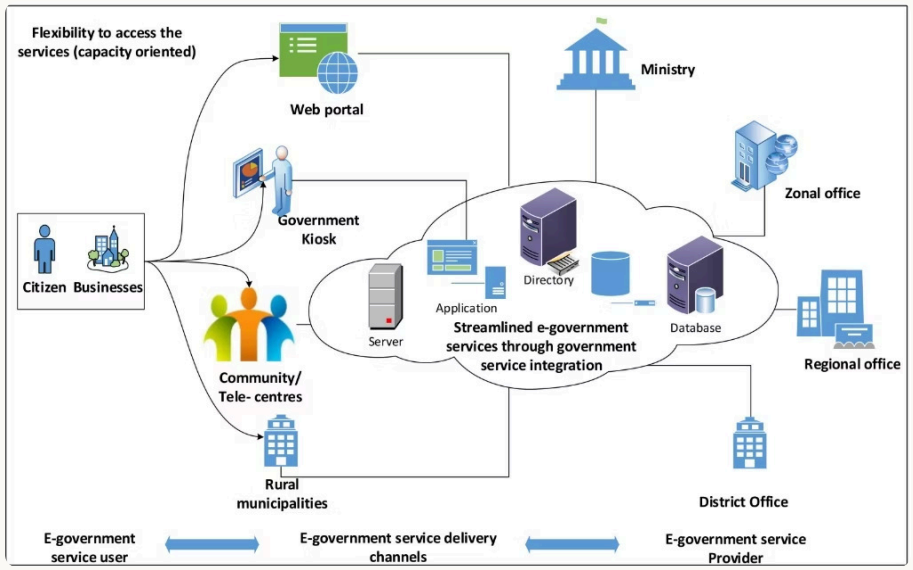
Rail Cloud hosts **NIVARAN**, a portal resolving employee grievances with an improved, responsive user experience.

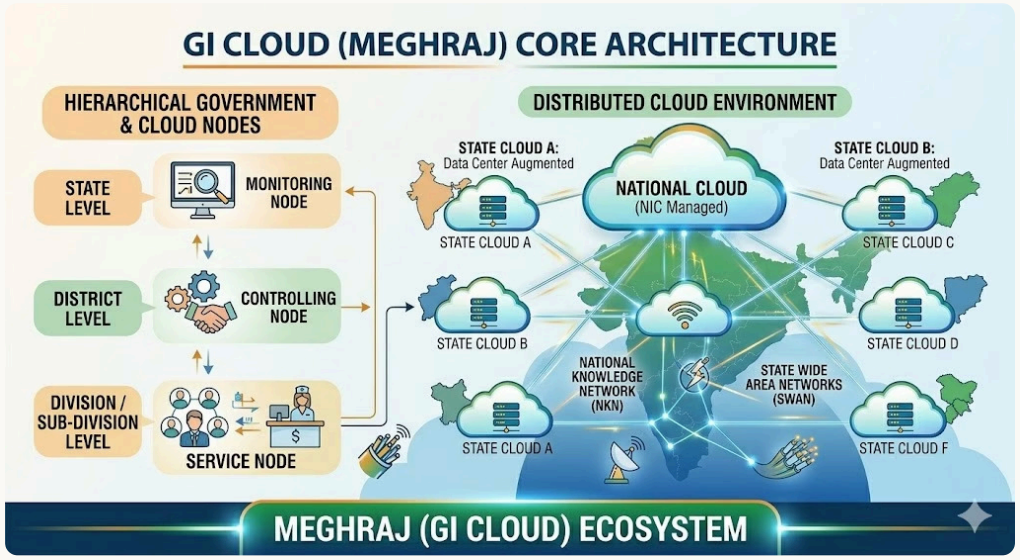
Rural Governance

e-Panchayat provides a high-level architecture to digitize and streamline village-level government operations.

Judicial & Public Safety

National Prisons Information Portal (NPIP) manages prisoner data and supports the **Kara Bazaar** inmate products marketplace.





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STRATEGIC BENEFITS

Major Advantages & Strategic Benefits

Economic Optimization

Pay-as-you-use model eliminates high upfront capital outlays for physical hardware procurement.

Operational Agility

The **e-Gov AppStore** offers reusable applications, dramatically reducing development and deployment cycles.

Technical Performance

Cloud systems deliver high availability with guaranteed uptimes approaching **99%**.

Interoperability

Standardized protocols enable application replication across states without duplicating infrastructure efforts.

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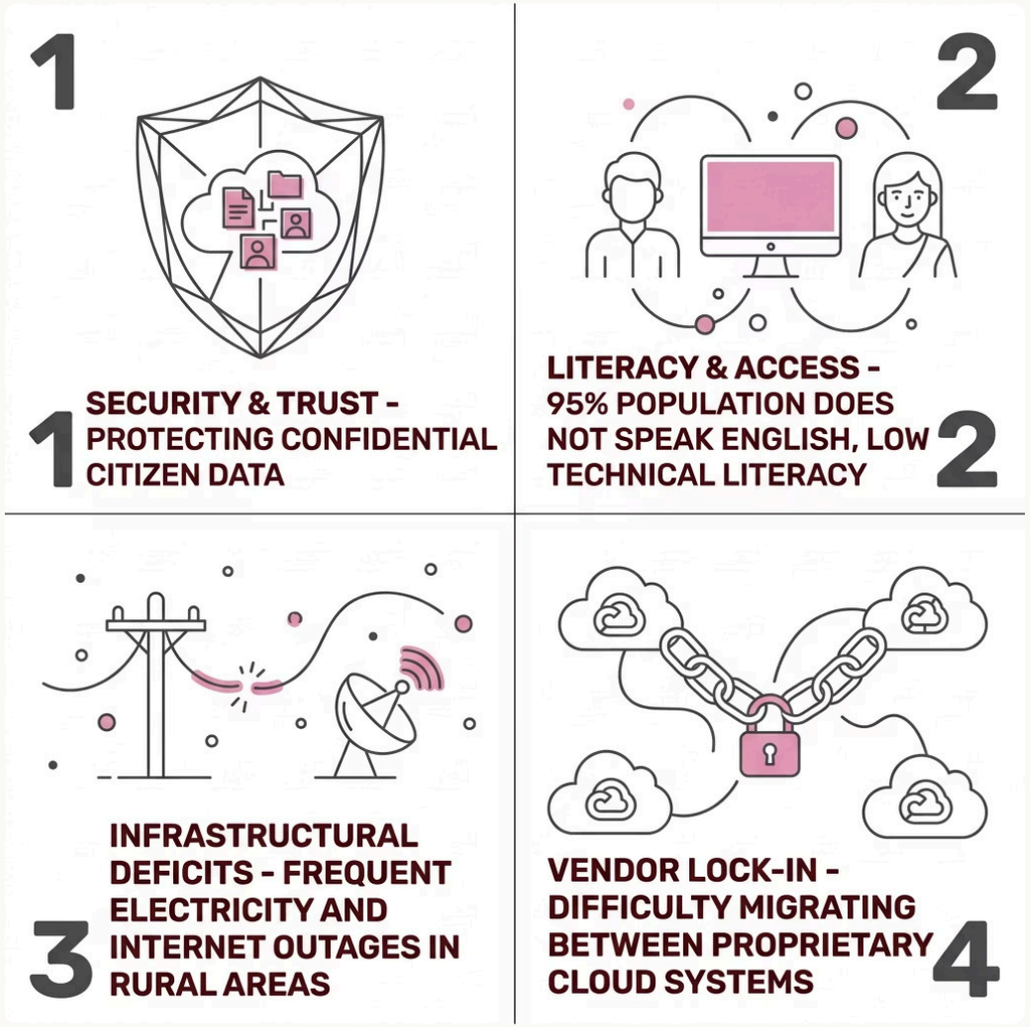
CHALLENGES

Disadvantages & Implementation Barriers



Key Barriers to Adoption

- **Security & Trust:** Protecting confidential, personal, and sensitive citizen data remains the foremost challenge.
- **Literacy & Access:** With **95%** of the population not speaking English and low technical literacy, usage barriers are significant.
- **Infrastructural Deficits:** Frequent electricity and internet outages in rural areas delay effective cloud service deployment.
- **Vendor Lock-in:** Concerns persist regarding the difficulty of migrating data between proprietary systems or switching providers.

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RECENT DEVELOPMENTS

Recent Developments & Policy Shifts

800+

Service Offerings

Grew from 80 in 2015 to over 800 by 2019 — a 10x expansion.

1.26L

Gram Panchayats

Connected via BharatNet optical fiber for rural cloud access.

2017

Data Residency Year

Mandatory guidelines requiring all government data to reside within Indian borders.

Private Sector Integration

Major Cloud Service Providers — **AWS, Microsoft, and IBM** — have been empanelled to support government projects, expanding capacity and expertise.

Data Sovereignty Mandate

The 2017 data residency guidelines ensure that all government data and services are hosted strictly within Indian borders, reinforcing national digital sovereignty.

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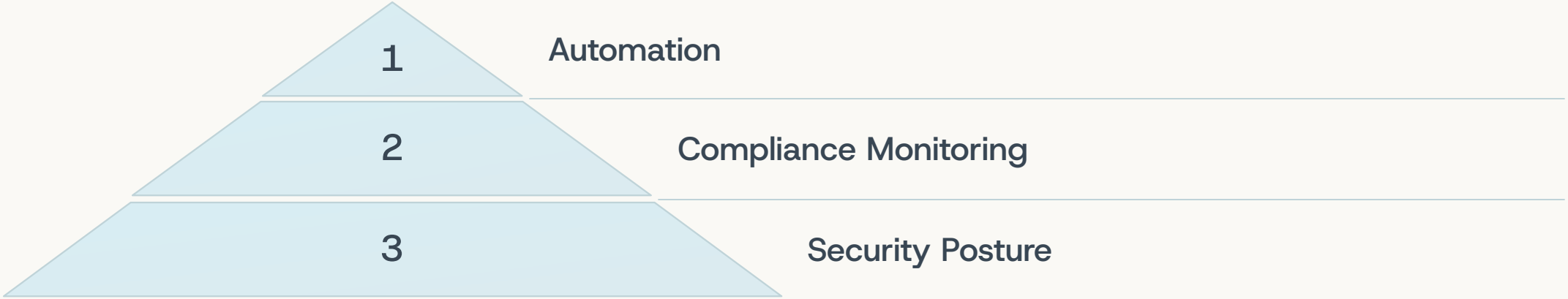
CONCLUSION

Conclusion — The Three Pillars for Success

Sustainable cloud governance in India rests on three interdependent pillars that together ensure security, compliance, and operational resilience.



Security Posture

Regular infrastructure snapshots to identify and remediate configuration errors before exploitation.

Compliance Monitoring

Real-time alerts for changes that might compromise regulatory or policy compliance.

Automation

Automated tools to remediate misconfigurations proactively, preventing security breaches at scale.

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